

Skills Summary

Graphing Basic Polar Curves from Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluate a polar equation at an angle

Idea: A polar equation $r = f(\theta)$ is a rule: feed it an angle, get a signed distance from the pole.

- The point is (r, θ) : go out r along the ray at angle θ .
- **Negative r is legal.** Plot $|r|$ in the direction *opposite* θ ; that is exactly how inner loops are drawn.
- Exact values only: $\cos 60^\circ = \frac{1}{2}$, $\cos 120^\circ = -\frac{1}{2}$, $\cos 180^\circ = -1$.

Example: For $r = 3 \cos \theta$, find r at $\theta = 60^\circ$.

Step 1. A single angle is given and a distance is wanted, so this is plain substitution — no equation to solve.

Step 2. $r = 3 \cos 60^\circ = 3 \cdot \frac{1}{2} \Rightarrow \frac{3}{2}$

Try it: For $r = 4 \sin \theta$, find r at $\theta = 30^\circ$.

check: $r = 2$

Skill 2 — Pole crossings and maximum radius

Two questions that build the picture:

- **Zeros.** Solve $f(\theta) = 0$ on $0^\circ \leq \theta < 360^\circ$. Each solution is an angle at which the curve passes through the pole — the boundary between petals, a cardioid's cusp, or the ends of an inner loop.
- **Maximum r .** Solve $f(\theta) = r_{\max}$. Those angles point at the petal tips, and for a circle or limaçon they give the direction the curve is offset from the pole.
- With $\sin(k\theta)$ or $\cos(k\theta)$, solve for $k\theta$ first, list every value up to $k \cdot 360^\circ$, and only then divide by k — dividing too early loses solutions.

Example: Find every θ in $[0^\circ, 360^\circ)$ with $3 \sin(2\theta) = 0$.

Step 1. Asking where the rose returns to the pole means solving $r = 0$; the doubled angle means the window for 2θ is twice as wide.

Step 2. $\sin(2\theta) = 0$ gives $2\theta = 0^\circ, 180^\circ, 360^\circ, 540^\circ$; halving each gives four angles. $\Rightarrow \theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ$

Try it: For $r = 5 \cos \theta$, solve $5 \cos \theta = 5$ on $[0^\circ, 360^\circ)$.

check: $\theta = 0^\circ$

Skill 3 — Identify the family, then sketch it

Recognise the form before plotting:

- $r = a \cos \theta$ or $a \sin \theta$: a **circle** through the pole, offset along the 0° or 90° ray.
- $r = a \sin(k\theta)$ or $a \cos(k\theta)$: a **rose** — k petals when k is odd, $2k$ petals when k is even, each of length $|a|$.
- $r = a \pm b \cos \theta$: a **limacon**. Equal a and b gives a cardioid (cusp); $a/b < 1$ gives an inner loop; $a/b > 1$ gives no loop.
- Then sketch from four values: $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ$, plus the zeros and the maximum.

Example: Sketch $r = 2 + 2 \cos \theta$. Start with its greatest radius.

Step 1. Constant and coefficient are equal, so this is a cardioid; the sketch is anchored by the largest radius, which occurs where $\cos \theta$ is largest.

Step 2. $\cos \theta$ peaks at $\theta = 0^\circ$, giving $r = 2 + 2(1)$; the curve then shrinks to 0 at 180° , the cusp. $\Rightarrow$ 4

Try it: For $r = 3 + 3 \sin \theta$, what is the greatest value of r ?

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